The 15:50 Candle: A Pre-Registered Study of an End-of-Day Signal in 0DTE SPX Options

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Abstract

We test whether a one-minute candle near the close predicts the S&P 500 index (SPX) into the close, and whether the prediction pays after the cost of trading same-day-expiry (0DTE) index options. Every confirmatory test was pre-registered. A scan of 41 candle features on 2013–2020 selected a 15:50 candle whose body spans at least 60% of its range; on 2021–2026 it was right on 57.5% of 644 trades. An SPXW option 10 points in the money, bought at the quoted ask and held to settlement, earned $67 a trade ($t = 2.40$), but the same trades lost money before 2021, when spreads were about five times wider, and limit orders at the mid did not help because they missed the best days. Twenty-six pre-registered stop and take-profit rules, tested at real bids, all earned less than holding on 2021–2023. The result depends on where the candle is read: from the index itself the rule missed its criterion ($t = 1.53$). In a pre-registered test with the exchanges' own messages, the candle pointed the same way as the closing-auction imbalance published at 15:50 on 73% of signal days but was not more accurate on those days (56% against 62%).

JEL classification: G13, G14. *Keywords:* 0DTE options; S&P 500 index options; closing auction; intraday predictability; bid-ask spread; pre-registration.

1 Introduction

At 15:50 ET the closing auction of the U.S. equity market comes into view. At that minute the New York Stock Exchange publishes its regulatory closing imbalance and Nasdaq begins to disseminate an early imbalance indicator for the close, and from then on the NYSE accepts new market-on-close and limit-on-close orders only on the side that offsets the published imbalance.[1] An option on the S&P 500 index that expires that day and is bought at 15:51 settles in cash on the official close nine minutes later, so its payoff depends on little more than the direction and size of the last move of the session. If the minute in which the market first sees the closing imbalance carries information about the close, same-day (0DTE) index options are the most direct way to trade it.

It is also a setting in which it is easy to be fooled. Many plausible candle rules, scored on enough history, produce an apparent winner by chance, and the winner's edge shrinks out of sample (Harvey et al., 2016). A backtest that enters at the close of the signal candle uses information that was not yet available, and a rule that is right 54% of the time can still lose money once the bid-ask spread is paid. We therefore pre-registered every confirmatory test (Nosek et al., 2018): what would be computed, on which data, and what would count as success. Each plan was hashed and

*This version: November 2026. Every confirmatory test was pre-registered: each plan was hashed and time-stamped before the data it governs was read, and the tests that failed are reported with those that passed. An Internet Appendix, the long working-paper version of this study, reports every study in full. A statement on data availability is at the end of the paper. This paper is research, not investment advice.

[1]See the NYSE fact sheet *NYSE Opening and Closing Auctions* and Nasdaq's *Closing Cross* FAQ. The NYSE moved its cut-off and imbalance publication from 15:45 to 15:50 on 1 April 2019, under a rule change the SEC approved on 31 January 2019 (Release No. 34-85021). Nasdaq moved the other way: in the fourth quarter of 2018 its cut-off and the first dissemination of its Net Order Imbalance Indicator moved from 15:50 to 15:55 (Release No. 34-84454), and it later added an abbreviated Early Order Imbalance Indicator from 15:50, scheduled for the fourth quarter of 2019 (Release No. 34-87179).

time-stamped before the data it governs was read, and every study is reported, including those that failed.

The paper proceeds in four steps. A scan of 41 candle features selects a signal on 2013–2020 and tests it on 2021–2026 index moves. Real OPRA quotes then price the trade, tuning on 2013–2020 and confirming once on 2021–2026. Trade bars and quote snapshots describe what execution costs. Finally, because the backtests read the signal from a contract for difference (CFD) that tracks the index, two further studies read the same candle from the SPX index and from SPY, and a last search compares the 15:50 candle with the other closing minutes.

Our main findings are as follows. First, one candle passes: the 15:50 one-minute candle with a *strong body*, traded in its own direction, predicts the move to the close out of sample. It is a sharper form of an effect that was already known, and it has weakened since 2023. Second, a deep in-the-money option bought at the quoted ask and held to settlement earned +\$67.46 per contract on 2021–2026 ($t = 2.40$), about two-thirds of what a calibrated pricing model implied. Third, the bid-ask spread decides whether the trade pays. The same trades lost money at the ask before 2021, a limit order at the mid does not recover the spread because it fills mainly when the trade is going wrong, and a one-minute delay costs a sixth of the edge. Fourth, the result depends on the source of the candle. Read from the SPX index candle, under a plan locked before any index candle was joined with a price, the rule made +\$39.44 a trade ($t = 1.53$) and missed its criterion; read from the SPY candle it made +\$71.74 ($t = 2.35$) and met a retrospective one. On all three sources the gain came on sessions on which every candle showed a strong body. Fifth, nothing tried afterwards improved on the rule, and a search over 7,128 settings of candle source, minute, cut, direction and strike chose the 15:50 candle again, although its choice was not clearly beyond luck ($p = 0.051$). Sixth, the candle pointed the same way as the closing-auction imbalance published at 15:50 on 73% of signal days, but in a pre-registered test it was not more accurate on those days (56% against 62%), so the auction explanation we propose was not confirmed, although the test could detect only a large difference.

The paper relates to four literatures. The first documents market intraday momentum. Gao et al. (2018) find that the first half-hour return predicts the last half hour of the S&P 500, and Baltussen et al. (2021) find the same pattern across futures markets and link it to the hedging of option dealers who are short gamma. The 15:50 candle lies inside that last half hour, yet in our data neither momentum nor a dealer-gamma proxy explains it, or improves it. The second studies the closing auction. Bogousslavsky and Muravyev (2023) show that closing-auction volume is tied to passive and ETF ownership and that auction imbalance information contributes to price discovery at the close; our signal sits in the first minute after that information is published, and we test whether its accuracy depends on agreeing with it. The third concerns 0DTE options. Beckmeyer et al. (2023) document that retail traders lose money in them, and Dim et al. (2024) find little evidence that 0DTE gamma or trading volume amplifies index moves. We show, at real quotes, how the spread determines whether a genuine short-horizon prediction can be monetised with these contracts. The fourth is data mining and pre-registration in return predictability (Harvey et al., 2016; Nosek et al., 2018). Our studies show how a plan, hash and ledger protocol can govern an evolving research programme, including the reporting of the tests that failed.

This is research, not a recommendation to trade. The holdout pass is nominal rather than blind, the real-price confirmation reuses index outcomes that had already been seen, the execution results are descriptive proxies, the confirmed rule reads its candle from a CFD rather than from the index, and the SPY candle's pass is retrospective. Section 2 describes the data and the design, Section 3 the signal, Sections 4 and 5 the trade at real prices and its costs, Section 6 the source of the candle and Section 7 the search for a better rule. Section 8 discusses why 15:50 might matter and tests one explanation, and Section 9 concludes.

2 Data and research design

2.1 Data

Table 1 lists the data sources.

Table 1: Data sources.

The index sample runs from 2013-02-28 to 2026-10-20 and the OPRA SPXW sample from 2013-04-01 to 2026-10-20. Robinhood one-minute SPX and SPXW bars, used to calibrate the pricing model and for the forward record, begin on 2026-09-21.

Source	Content	Period	Used for
Dukascopy	US500 CFD one-minute bars, regular trading hours	2013–2026	Signals; index outcomes
Databento OPRA	SPXW best bid and offer at one-minute marks, 15:00–16:01 ET, the 25 strikes nearest the money, calls and puts	2013–2026	Option prices
Databento OPRA	SPXW one-minute trade bars for the same contracts	2013–2026	Execution
Databento OPRA	Daily open interest, all SPX and SPXW options	2013–2026	Gamma proxy
Databento OPRA	XSP and SPY quotes	2021–2026	Smaller contracts
Cboe	Official SPX daily close	2013–2026	Option settlement
CNBC	SPX index one-minute bars (archive)	2021–2026	Index candle (Section 6)
CNBC	SPY one-minute bars (archive; traded prices, not dividend-adjusted)	2021–2026	SPY candle (Section 6)
Databento	NYSE and Nasdaq closing-auction imbalance messages, 15:50–16:00 ET on signal days, for SPY's 50 largest holdings (SEC N-PORT reports)	2021–2026	Imbalance test (Section 8.2)

Index. Signals and index outcomes come from Dukascopy's US500 CFD one-minute bars during regular trading hours, restricted to a validated set of full sessions. Discovery uses 2013–2020 (1,861 sessions) and the holdout 2021–2026 (1,446 sessions). The CFD stands in for SPX, and index outcomes are measured to the close of the 15:59 bar rather than to the official settlement. Section 6 reads the signal candle instead from archives of SPX and SPY one-minute bars.

Options. Real option prices come from Databento's OPRA feed: the consolidated best bid and offer at one-minute marks for SPXW contracts expiring that day, from 15:00 to 16:01 ET, for the 25 strikes within 60 points of the at-the-money strike at 15:51. A quote stamped HH:MM:00 is the quote at the open of that minute. One-minute trade bars for the same contracts support the execution analysis, and daily open interest on every SPX and SPXW option supports a dealer-gamma proxy. Of the 3,287 index days, 2,149 had an SPXW expiry that day; the rest are excluded from every option study. Options settle at intrinsic value on Cboe's official SPX close.

2.2 The signal and the trade

Let o, h, l, c be the open, high, low and close of a one-minute bar. The bar has a *strong body* if $h > l$ and

$$|c - o| \geq 0.6\,(h - l), \tag{1}$$

and its direction is the sign of $c - o$. The signal is a strong body on the 15:50 bar. Because that bar is complete only at 15:50:59, the trade enters at the open of the 15:51 bar and, in the index tests, is scored on the move to the close of the 15:59 bar.

In the option studies, let S be the index level at entry and $K_{\text{ATM}} = 5\lfloor S/5 + 1/2 \rfloor$ the nearest listed strike. An offset of j points, positive meaning in the money in the trade's direction, buys the

call struck at $K_{\text{ATM}} - j$ after an up signal and the put struck at $K_{\text{ATM}} + j$ after a down signal. We consider nine offsets, from 20 points in the money (ITM20) to 20 points out of the money (OTM20) in steps of five. One contract (multiplier 100) is bought per signal and held to cash settlement, and each side pays a fee of \$0.09 per contract, with \$0.74 as a sensitivity. We call the 15:50 strong-body signal traded at ITM10 the *rule*.

2.3 Pre-registration

Each confirmatory study was specified in a written plan before the data it governs was read. The plan's SHA-256 hash and a UTC timestamp were appended to a ledger, and each study's code recomputes the hash at start-up and refuses to read real data unless it matches a ledger entry. Each plan lists what was already known when it was written, and any departure from a plan is reported as a labelled deviation; no result was re-run under a changed rule. Table 2 lists every study, its primary criterion and its outcome. Analyses that were not pre-registered are labelled *descriptive*; they inform the discussion but decide nothing.

Table 2: The pre-registered studies.

Registration dates are UTC. The strike-ladder and early-exit studies priced trades with a calibrated model, before real quotes were available; they and the forward log are reported in full in the Internet Appendix. The limit-order and fixed-minute sale analyses of Section 5 were not pre-registered.

Study	Section	Registered	Primary criterion	Outcome
Candle scan	3	2026-10-21	Holm-corrected test of each frozen candidate on 2021–2026, and an accuracy bar	One of three candidates passes
Strike ladder	4.3	2026-11-08	Nine strikes ranked under a calibrated model	Deep in the money most robust
Early exits	7.3	2026-11-09	A rule beats holding with $t > 2.64$, robust across model variants	No rule passes
Real option prices	4	2026-11-09	Switch only if a new setting leads by $t \geq 2.0$; confirm with $t \geq 2.0$	Rule kept and confirmed
Re-tuning	7.1	2026-11-09	Luck diagnostic $p \leq 0.05$, then $t \geq 2.0$ on 2024–2026	Pick fails the luck diagnostic
Intraday momentum	7.3	2026-11-09	Slope $t \geq 2.0$; filters $p \leq 0.0125$; gamma $p \leq 0.025$	Nothing supported
XSP and SPY	7.3	2026-11-09	Positive mean with $t \geq 2.0$ on 2021–2026	Neither confirmed
Index candle	6	2026-11-10	Rule read from the index candle: positive mean per day with $t \geq 2.0$	Criterion not met
SPY candle	6	2026-11-10	Rule read from the SPY candle: positive mean per day with $t \geq 2.0$	Retrospective criterion met
Candle search	7.2	2026-11-10	Best of 7,128 settings on 2021–2023, tested on 2024–2026 only if not at 15:50 and luck diagnostic $p \leq 0.05$	Pick at 15:50; no test
Closing imbalance	8.2	2026-11-10	Hit rate higher on days the candle agrees with the 15:50 imbalance: $d > 0$ with $t \geq 2.0$	Not confirmed
Real-time exits	5.3	2026-11-10	An exit rule beats holding in both halves of 2021–2023 and has luck diagnostic $p \leq 0.05$; then $t \geq 2.0$ on 2024–2026	Stopped at the coverage gate
Real-time exits, revised	5.3	2026-11-11	The same, for the 26 rules that read only the option's bid	No eligible rule; no test
Forward options log	7.3	2026-10-30	Running record of real option outcomes	Four priced trades so far

2.4 Statistical inference

We report mean P&L per trade and per day, where the daily series takes the trade's P&L on a signal day and zero otherwise. Unless stated otherwise, t-statistics use Newey–West standard errors with five lags on the daily series (Newey and West, 1987). The scan tests its candidates one-sided with a Holm correction (Holm, 1979), and hit-rate intervals are Wilson intervals (Wilson, 1927).

Studies that choose one setting from a large grid also report a *luck diagnostic*. The signal series is shifted circularly against the outcome days, by every shift from 21 days to $n-21$ in steps of three. A shift keeps the signal's own structure but breaks its link to the outcomes. The whole search is repeated for each shift and its best score kept. With S shifts, of which k produce a best score at least as good as the real one, $p = (1 + k)/(1 + S)$. The diagnostic asks whether the real winner beats the winner that unrelated signals would find. It is approximate, because expiry frequency, volatility and pricing change over the sample.

3 Does the 15:50 candle predict the close?

Design. We defined 41 features, each of which outputs up, down or no trade using only bars completed before the entry: the direction of every one-minute candle from 15:30 to 15:50; five candle shapes (strong body, hammer, shooting star, engulfing, big candle) on the 15:50 one-minute candle and on the 15:45–15:49 five-minute candle; five-minute directions; and four whole-day features. Features with $|t| \geq 2.5$ on 2013–2020 qualified, up to eight. The list of qualifying features was frozen and hashed before any 2021 bar was read. On 2021–2026 each candidate faced a one-sided test with a Holm correction at 5%. A candidate that passed was called *accurate* only if the lower bound of its 95% Wilson interval exceeded the hit rate at which buying the at-the-money option breaks even.

Table 3: The candle scan: frozen candidates on the 2021–2026 holdout.

Panel A gives hit rates in discovery (2013–2020) and on the holdout, with 95% Wilson intervals. The strong-body break-even is 49.5% at a \$0.09 fee (51.8% size-neutral); with 644 trades, "accurate" requires an observed hit rate of 53.4% (55.7% size-neutral). The P3 row is a reference and was not tested. Panel B prices the at-the-money option with the scan's cost model. By year from 2021 to 2026 the hit rate was 64.9, 61.0, 54.0, 53.8, 57.3 and 53.3%.

Panel A. The three candidates

Candidate	2013–20	2021–26 (95% CI)	trades	Holm p	predicts	accurate
15:50 strong body	54.1%	**57.5%** (53.6–61.2)	644	6.3×10^{-5}	yes	**yes, just**
15:50 direction, any body	52.2%	54.0% (51.4–56.6)	1,445	0.011	yes	no
Day so far (09:30–15:49)	52.0%	48.4% (45.8–51.0)	1,446	0.53	no	no
P3 reference (not tested)	*51.8%*	*53.6%* (51.1–56.2)	*1,442*	—	—	—

Panel B. The strong-body candle by subperiod (descriptive)

Strong body	trades	hit rate (95% CI)	t	option P&L / trade (t)
2021–2022	216	63.0% (56.3–69.1)	3.56	+\$80.7 (2.33)
2023–2026	428	54.7% (49.9–59.3)	2.27	+\$28.6 (1.11)

Results. Three features qualified (Table 3, Panel A). Two of them, both on the 15:50 candle, predicted the 2021–2026 move, but only the strong-body candle also cleared the accuracy bar. It was right on 57.5% of 644 trades, and its trades moved an average of 1.82 index points in the signal's direction over nine minutes, at today's index level. The third candidate, a whole-day feature, did not survive out of sample.

Robustness. Two checks argue against chance and against a generic intraday pattern. When the 41 features were shifted out of line with the 2013–2020 outcomes and the scan was re-run 304 times, about 1% of runs found as many qualifying features as the real scan, and none found one as strong as the best real feature. The same rules moved four hours earlier, entering at 11:51, showed nothing in either period.

A known effect in a sharper form. Both passing features sit on the 15:50 candle and agree with the direction of the move from 15:50 to 15:51 (an effect we call P3, found in earlier exploratory work) on 98% and 93% of days. P3's behaviour over 2013–2026 was known before the scan, so the holdout was not blind for these features. The fair reading is that a known effect survived a stricter test in a sharper form. On 15:50 candles without a strong body, trading the direction earned nothing.

Weaker since 2023. Most of the edge comes from 2021–2022 (Table 3, Panel B). In 2023–2026 the hit-rate interval includes 50%. At that effect size, a test with 80% power would need roughly 19 years of signals, so no single year can settle the question.

4 The trade at real option prices

4.1 Design

At the entry minute, the spot S is implied by put–call parity: it is the median of $K + C_{\text{mid}} - P_{\text{mid}}$ over the three strikes where the call and put mids are closest. The trade buys at the ask at the open of the entry minute and settles on the official close, so its P&L per contract is

$$\text{P\&L} = 100\,(\text{intrinsic value at the official close} - \text{ask}) - \text{fee}. \tag{2}$$

A trade with no implied spot or no valid quote is dropped and counted.

The study first tuned on 2013–2020 over a grid of 2,178 settings: the candle minute (any one-minute candle from 15:40 to 15:50, entering at the next minute's open), the body threshold (0.40 to 0.90 in steps of 0.05), the direction (with or against the candle) and the strike (ITM20 to OTM20). A setting's score is its mean P&L per day averaged with its neighbours at adjacent thresholds, which favours a plateau over a sharp peak, and a setting is eligible with at least 150 trades, at least 50 in each half of the period, and a positive mean in both halves. The best eligible setting would replace the rule only if its luck diagnostic gave $p \leq 0.05$ and its per-day lead over the rule had $t \geq 2.0$. The chosen rule is confirmed on 2021–2026 if its mean P&L per day is positive with $t \geq 2.0$. The plan states that the 2021–2026 index outcomes had already been used to test the signal, so the confirmation is fresh for option prices but not for the index moves.

Tuning was weak. Spreads were wide and many signal days could not be priced, so only two of the 2,178 settings were eligible. The better one, the 15:42 candle with a 60% body at the money, passed the luck diagnostic ($p = 0.034$), but its lead over the rule was not significant ($t = 1.56$), and the rule went forward unchanged. The rule itself lost money at the ask in this period.

4.2 Confirmation on 2021–2026

The rule was confirmed (Table 4). It earned +$29.87 a day with $t = 2.40$, against a bar of 2.0. The mean trade made +$67.46, slightly more than half of all trades were profitable, and the worst lost $2,210 on one contract. A five-day block bootstrap puts the mean between +$21 and +$115 a trade (90% interval): positive, but imprecisely measured. Figure 1 shows the full record. Three qualifications apply. The result confirms the rule at real prices but does not show that it is the best setting, since no other setting was evaluated on this period (Section 7 returns to that question). The index outcomes had already been seen. And the edge is weaker since 2023, when it is not significant on its own.

Table 4: Confirmation: the rule at real asks, 2021–2026.

The rule (15:50 candle, body at least 60%, with the candle, ITM10, held to the official close) at real asks, $0.09 fee, dollars per contract. t is Newey–West on the per-day series. Subperiods are descriptive. Mean per trade by year: 2021 +$175 (67 trades), 2022 +$71 (90), 2023 +$4 (113), 2024 +$20 (106), 2025 +$65 (117), 2026 +$121 (92).

Period	days	$/day	t	trades	$/trade	won	median	worst
2021–2026	**1,321**	**+29.87**	**2.40**	**585**	**+67.46**	**54%**	**+44.91**	**−2,210**
2021–2022	376	+48.21	2.01	157	+115.46	57%	+65.91	−1,361
2023–2026	945	+22.58	1.56	428	+49.85	53%	+35.91	−2,210

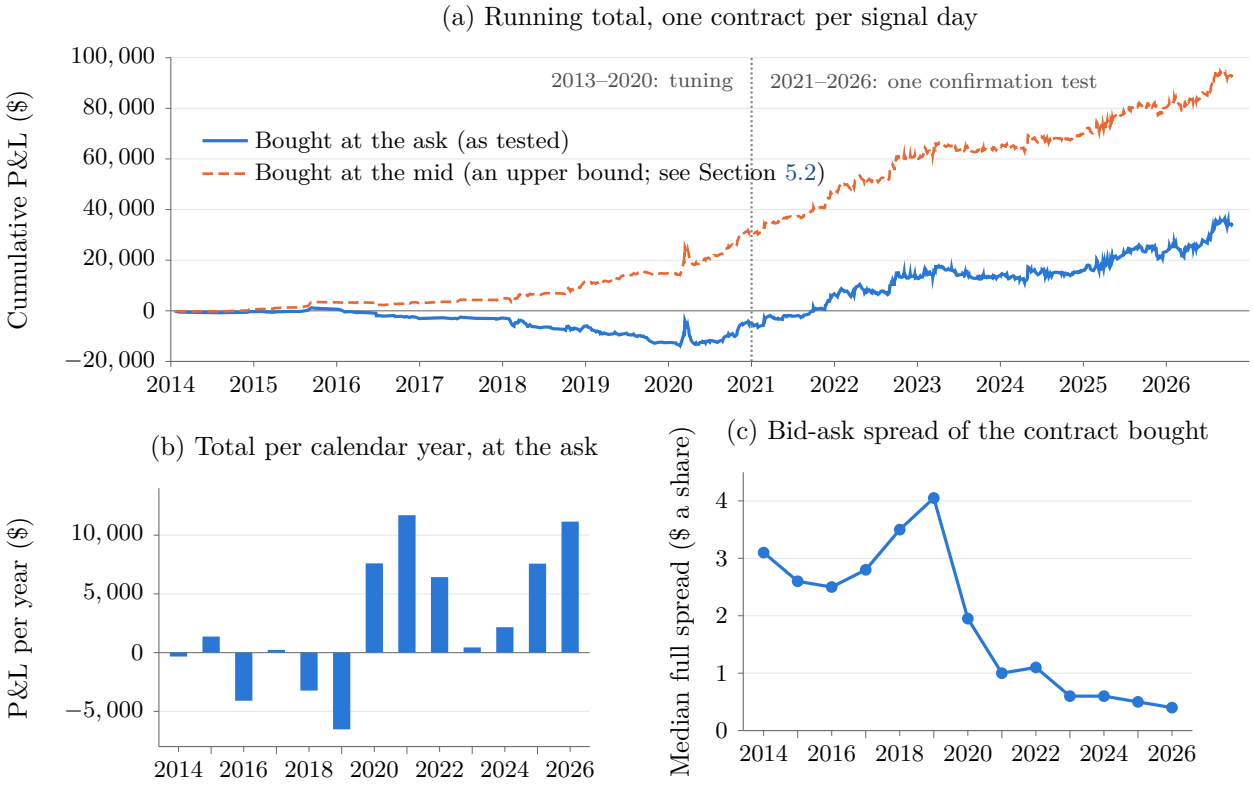


Figure 1: The rule (15:50 candle, body at least 60%, with the candle, ITM10, held to the official close) at real Databento OPRA prices, $0.09 fee, dollars per contract at each year's prices. (a) The 801 priced signal days, 2014–2026. At the ask the rule lost $4,997 over the 216 trades of 2013–2020 and made $39,463 over the 585 of 2021–2026. The dashed line buys at the mid and is only an upper bound: a limit order at the mid made about what paying the ask made, because it missed the best days (Section 5.2). (b) 2023 and 2024 were close to zero. (c) Median ask minus bid at entry. The narrowing spread is why the same signal lost money at the ask before 2021 and made money after it. Before 2016 SPXW had a same-day expiry only on some days, so the early years have few trades.

4.3 Strikes

Table 5: The signal at every strike, real asks, 2021–2026.

$0.09 fee, dollars per contract. Descriptive, not a second test: choosing a strike from this table would be choosing on the confirmation period.

Strike	trades	$/trade	won	median	worst	$/day	t
ITM20	586	+77.15	54%	+51.91	−2,670	+34.20	2.61
ITM15	584	+71.00	55%	+52.41	−2,330	+31.41	2.43
itm10	**585**	**+67.46**	**54%**	**+44.91**	**−2,210**	**+29.87**	**2.40**
ITM5	585	+53.90	50%	+2.91	−1,900	+23.87	2.07
ATM	583	+39.50	39%	−110.09	−1,640	+17.46	1.87
OTM5	585	+12.18	19%	−65.09	−1,390	+5.40	0.78
OTM10	581	+1.04	11%	−25.09	−1,180	+0.46	0.10
OTM15	566	−6.96	5%	−15.09	−1,000	−3.02	−1.03
OTM20	529	−10.02	3%	−10.09	−840	−4.19	−2.34

Before real quotes were available, a pre-registered study priced the holdout trades with a Bachelier model (Bachelier, 1900), calibrated on seven days of real broker prices, and found strikes 10 to 20 points in the money the most robust. At real prices the strikes rank the same way (Table 5). Deep in-the-money options carry little time value, so they keep nearly all of a small move; ITM10 needs about half the premium of ITM20. At the money the mean was positive but most trades lost, and ten or more points out of the money the option rarely paid. On the same days and contracts, the model implied +$100.70 a trade at ITM10 and the real trade made +$67.77. In the money the model priced the option well, with the real mid within about 2% of the model price, but the real ask sat further above the mid than the model's flat half-spread allowed.

5 Costs and execution

5.1 The bid-ask spread

Table 6: The rule bought at the ask and at the mid.

ITM10, $0.09 fee, dollars per contract. Rows at the mid are descriptive and decide nothing. The spread is ask minus bid per share at entry, the median over the priced trades. With a $0.74 fee the ask rows become −$23.79 and +$66.81 a trade ($t = 2.37$ in 2021–2026).

Period	bought at	trades	$/day	t	$/trade	won	median full spread
2013–2020	ask	216	−7.70	−0.47	−23.14	42%	$2.90
	mid	216	+48.19	2.87	+144.78	56%	
2021–2026	ask	585	+29.87	2.40	+67.46	54%	$0.60
	mid	585	+46.90	3.71	+105.90	56%	

Table 6 and Figure 1(c) contain the study's main lesson. At the mid, the same trades made money in both periods. At the ask they lost money in 2013–2020, when the median quoted spread was about five times its 2021–2026 level, and made money afterwards. Even since 2021, paying the ask rather than the mid costs about $39 a trade, more than a third of the edge at the mid. Whether the trade pays therefore depends on how the order is filled.

The ask is a price that was offered, not one that was paid. OPRA trade bars show that since 2021 the contract traded in the entry minute on 99% of signal days, against just over half the days before, and on 95% of those days the minute's lowest trade was at or below the 15:51 ask. A trade bar does not show the size available, the queue, or what a particular order would have paid, so the fills assumed here remain an approximation.

5.2 Limit orders and late entry

At the mid the trade made far more than at the ask, which suggests an easy gain. We simulated six order styles, fixed before any result was computed: paying the 15:51 ask (P0); a limit at the 15:51 mid for one minute, else no trade (P1); the same, but counting a trade at the limit as a fill (P1t); P1 followed by paying the 15:52 ask if unfilled (P2); a mid limit left working for three minutes (P3); and a limit halfway between the mid and the ask, then the 15:52 ask (P4). A limit counts as filled if a trade printed strictly below it during a minute it was working, or if the quote at the end of that minute offered the option at or below it. These are proxies: trade prints cannot show whether a resting order would have executed, and the order is assumed to be live with no delay.

Table 7: Order styles for the rule, 2021–2026.

ITM10, $0.09 fee, dollars per contract per priced signal day; an unfilled order counts as $0. The subperiods have 157 (2021–2022) and 428 (2023–2026) signal days. The t-statistics here treat signal days as independent, so P0's t is 2.29 rather than the per-day Newey–West 2.40 of Table 4. Descriptive.

	2021–2026, 585 priced signal days				minus P0, $/day	
Order style	filled	$/day	t	minus P0 (t)	2021–22	2023–26
P0 pay the 15:51 ask	585 (100%)	+67.46	2.29	—	—	—
P1 mid, 1 minute, else skip	489 (84%)	+64.57	2.37	−2.89 (−0.24)	−22.31	+4.23
P1t as P1, touch counts	501 (86%)	+61.04	2.23	−6.42 (−0.56)	−22.31	−0.59
P2 mid, 1 minute, then 15:52 ask	585 (100%)	+69.63	2.38	+2.17 (0.55)	+13.06	−1.82
P3 mid, 3 minutes, else skip	534 (91%)	+64.89	2.31	−2.57 (−0.26)	+6.44	−5.87
P4 halfway, 1 minute, then 15:52 ask	585 (100%)	+62.23	2.13	−5.23 (−1.63)	−2.36	−6.29

Table 8: Why the mid limit does not help, and what a late entry costs.

The rule, ITM10, $0.09 fee, dollars per contract. Panel B uses the days with a quote at every one of these minutes: 216 (2013–2020), 585 (2021–2026), 157 (2021–2022) and 428 (2023–2026), which is every priced day. Descriptive.

Panel A. The same entries at the ask, split by whether P1's mid limit filled

Period	P1	days	$/trade at the ask ($t$)	won
2021–2026	filled	489	+42.52 (1.31)	52%
	not filled	**96**	**+194.49 (2.80)**	**61%**
2013–2020	filled	69	+44.11 (0.37)	46%
	not filled	147	−54.70 (−1.42)	40%

Panel B. The same contract bought at a later minute's ask, $/trade

Buy at	2013–2020	2021–2026 (t)	2021–2022	2023–2026
15:51	−23.14	+67.46 (2.29)	+115.46	+49.85
15:52	−68.04	+55.75 (2.02)	+111.07	+35.46
15:53	−58.09	+57.89 (2.16)	+128.30	+32.06
15:55	−128.97	+23.01 (1.11)	+51.10	+12.71

No order style beat paying the ask. On 2021–2026 every style earned roughly what paying the ask earned, and every difference was within noise (Table 7). The reason is adverse selection (Table 8, Panel A). A limit at the mid fills when someone sells below the mid, which happens most when the option is getting cheaper, that is, when the trade is going wrong. When the trade works at once, the price runs away and the limit is left unfilled. The days on which the mid limit filled earned well below the average at the ask, and the days it missed earned far more. The limit saves money on the days it fills and misses the best days, and the two effects roughly cancel. Before 2021, with spreads about five times wider, the mid limit did help, but on a small sample.

Being late costs money. Bought at a later minute's ask, the same contract earned less the later the entry (Table 8, Panel B): about $12 a trade less at 15:52, and about two-thirds less at 15:55. The results assume entry at 15:51:00 exactly, while a person reacting to the candle trades seconds later.

5.3 Selling before the close

Selling early does not help. Each of the 585 confirmation trades, bought at the 15:51 ask, was also sold at the bid at the start of each later minute, paying the fee twice. Every fixed exit minute from 15:52 to 15:59 earned less than holding to settlement, by $79 to $136 a trade, and no year favoured an early exit. The pre-registered model study of stop-loss and take-profit rules reached the same conclusion (Section 7.3).

Real-time exit rules. A pre-registered study tested rules that watch the option's own bid at each minute from 15:52 to 15:58 and sell at the next minute's bid: 5 trailing stops (sell when the bid falls 10–50% below its peak since entry), 9 profit locks (once the bid has risen 10–30% above the price paid, sell if it gives back a third, two-thirds or all of that gain) and 12 take-profits (sell once the bid is 10–30% above the price paid, from 15:52, 15:54, 15:56 or 15:58). On the 270 trades of 2021–2023, a rule had to sell early on at least 25 trades and beat holding in each half of the period; the best such rule would then have faced a luck diagnostic and one test on 2024–2026. A first plan, which also included 18 rules that trail the index level implied by put–call parity, stopped at its coverage gate before any rule was computed, because that level was often missing in the last minutes; a revised plan, written after that stop, kept the 26 bid-based rules, whose quotes were complete. No rule qualified, so nothing was tested: all 26 earned less than holding on 2021–2023, by $9 to $120 a trade. Take-profits did worst, $49 to $107 a trade below holding, because they sell the trades that would have kept rising and do nothing for the losers. Looked at afterwards, 2024–2026 agrees: take-profits trailed holding by $39 to $108 a trade, and the stops were within $31 of holding, none with $|t|$ above 1.6.

6 Which candle?

Every signal so far was read from the CFD candle, while a trader reads the SPX index candle. The two track each other closely in level and direction, but a candle's body-to-range ratio depends on its shape, and the shapes differ. On 36 sessions in the autumn of 2026 with both series, the index candle showed a strong body at 15:50 on 20 days and the CFD candle on 14, all of them among the index candle's 20; over 2021–2026 the two ratios have a correlation of 0.78, and the CFD's 0.60 corresponds to about 0.69 on the index candle. Two pre-registered studies therefore read the 15:50 candle from archived one-minute bars, the SPX index candle first and the SPY candle second, and traded the same SPXW option at the prices of Section 4. The option outcomes of these days were already known, so both studies are retrospective checks of the signal's source, not independent confirmations.

Design. The archives, both from CNBC, hold a valid 15:50 bar on 1,310 (index) and 1,312 (SPY) of the 1,322 sessions from 2021-01-01 to 2026-10-20 that have option quotes and an official close. Checks made before each plan was locked, on price levels only, confirmed the time stamps; for SPY they also showed that the bars are traded prices, not dividend-adjusted, and that a second vendor's SPY bars agreed to the cent on 14 of 22 sessions compared, with differences of up to $0.02 elsewhere. The signal is the same body test, computed in whole cents, and the trade is unchanged: ITM10 in the candle's direction at the 15:51:00 ask, held to settlement. Each primary line asks for a positive mean per day with Newey–West $t \geq 2.0$; secondary lines also require a Holm-adjusted $p \leq 0.05$. The SPY study's second line was chosen after the agreement of two candles had been

seen to do well, and its lines are retrospective numerical criteria.

Table 9: The rule read from three candle sources, 2021–2026.

One ITM10 contract at the 15:51 ask, held to settlement, 2021-01-01 to 2026-10-20, dollars per contract. Panel A: the first row is the confirmation of Table 4; the index and SPY rows are the registered lines of the two source studies, whose primary lines are the first of each block. Per day is the mean of the daily series (0 on days outside the cell); t is Newey–West, lag 5; Holm p is adjusted across each study's secondary lines. On the same days as the index study the CFD rule made +$67.24 a trade (583 trades), and on the same days as the SPY study +$68.84 (582). Panel B: ITM10 in the direction of the first signalling candle in the order SPY, index, CFD; 1,300 sessions with all three candles; t is a simple t-statistic; no session had signalling candles in opposite directions. Panel B is descriptive.

Panel A. The registered lines

Signal	Trades	$/trade	At mid	Won	$/day	t	Holm p	Reading
CFD body ≥ 60% (the rule)	585	+$67.46	+$105.90	54%	+$29.87	+2.40		confirmed
Index body ≥ 60%	710	+$39.44	+$78.14	52%	+$21.39	+1.53		not met
Index body ≥ 70%	579	+$53.73	+$91.20	53%	+$23.75	+1.86	0.094	insufficient
Index body 60–70%	131	−$23.69	+$20.44	49%	−$2.37	−0.41	0.685	insufficient
SPY body ≥ 60%	523	+$71.74	+$108.51	53%	+$28.62	+2.35		met, retrospective
SPY and index both ≥ 60%	504	+$80.44	+$117.34	54%	+$31.21	+2.56	0.0053	met, retrospective
SPY body ≥ 70%	369	+$103.40	+$137.67	55%	+$29.08	+2.84	0.0045	met, retrospective

Panel B. Which of the three candles show a strong body

SPY	Index	CFD	Days	Trades	Ask	Mid	Won	t
yes	yes	yes	436	435	+$112.78	+$150.18	56%	+3.11
yes	yes	–	69	69	−$123.42	−$89.66	36%	−1.73
yes	–	yes	9	9	−$118.42	−$91.20	33%	−0.80
yes	–	–	6	6	−$401.42	−$378.09	33%	−1.25
–	yes	yes	82	82	−$77.58	−$28.55	44%	−1.20
–	yes	–	121	121	−$46.04	−$6.66	50%	−0.82
–	–	yes	54	54	−$33.83	−$2.53	54%	−0.49
–	–	–	523	0				

The index candle. Read from the index candle, the rule made +$39.44 a trade, against +$67.24 from the CFD candle on the same days, and its daily series fell short of the criterion ($t = 1.53$; Table 9, Panel A). The band of 70% or more came closer and also fell short, and the 60–70% band showed no edge. The directional effect is present on the index candle: after index candles of 70% or more the index moved +2.63 basis points in the candle's direction ($t = 4.55$), and after candles of 60–70% it did not. It is the option trade's result that falls short, because reading the index candle adds days without the effect. The 190 trades taken on days on which only the index candle showed a strong body lost $74.14 a trade, and 17% of index candles of 70% or more were below 60% on the CFD.

The SPY candle. Read from the SPY candle, the rule made +$71.74 a trade ($t = 2.35$), about what the CFD candle made on the same days, and all three registered lines met their retrospective criteria. The band of 70% or more was the strongest line and was positive in each of the six years; the 60–70% band made nothing, as on the other two sources. A strong SPY candle is nearly always a strong CFD candle: 94% of SPY candles of 70% or more were CFD signals.

Three candles. Across the three sources the pattern is the same (Table 9, Panel B). The gain was confined to the 435 sessions on which all three candles showed a strong body, and every other combination lost on average, including the index and CFD candles signalling without SPY. No single row other than the first is significant on its own, but together they say that a strong body on

only some of the candles is not the signal. One explanation, which we did not test, is that the CFD and SPY follow the futures, which move first, while the index is computed from the last trades in 500 stocks and can show a strong candle while it catches up. Because SPY was the third source tried on the same outcomes, these results are leads for a forward record rather than confirmations.

7 The search for a better rule

7.1 Re-tuning on recent data

Tuning on 2013–2020 could not tell nearby settings apart, so a further pre-registered study repeated the search over the same 2,178 settings on 2021–2023, the first three years of tight spreads and, from 2022, daily SPXW expiries, and would test its pick once on 2024–2026. The pick had to lie outside the rule's own family, pass the luck diagnostic and lead the rule on the tuning years; tuning and testing were separate runs, and the tuning step could not read any date after 2023. Of 119 eligible settings, the best was the 15:50 candle with a body of at least 80%, traded with the candle at ITM20. It failed the luck diagnostic ($p = 0.21$): about one shift in five found a best score at least as good with signals unrelated to the days. No candidate advanced and the rule stayed. Computed only after the decision was recorded, the pick earned +$18.88 a day on 2024–2026 against the rule's +$29.97, which is what a lucky winner looks like.

The body threshold itself cannot be tuned finely. In 2021–2026, the 15:50 candle traded in its direction earned −$37 a trade at the ask when its body was below 60% of the range and +$67 when it was at least 60%, and the index move shows the same break in 2013–2020. Above 60%, per-day results are flat: the bootstrap interval for the difference between thresholds of 0.70 and 0.60 runs from −$10.7 to +$8.8 a day, and a walk-forward choice of threshold earned less than a fixed 0.60. Within the signal, a larger ratio was not followed by a larger gain. Across all days the ratio's correlation with what followed was significant but faint, explaining at most 0.7% of the day-to-day variation, and among signal days it could not be distinguished from zero. The ratio behaves as a threshold near 60%, not as a measure of strength.

7.2 Other closing minutes

A last pre-registered study asked whether another closing candle does better. It covered 7,128 settings: three candle sources (the index candle, the SPY candle, and the two agreeing in direction), the eleven one-minute candles from 15:45 to 15:55, twelve body cuts from 40% to 95%, trading with or against the candle, and nine strikes from ITM20 to OTM20. Each trade buys at the ask in the snapshot that opens the next minute and holds to settlement. Settings were scored on 2021–2023 as before, and 387 were eligible. The best one would be tested once on 2024–2026 only if it was not a 15:50 setting, because those had already been examined on the later years, and only if its luck diagnostic, repeated over 195 shifts, gave $p \leq 0.05$. Before any statistic, the code reproduced the trades of the two candle studies exactly. It also passed synthetic self-tests and a full trial on synthetic data.

The search chose a 15:50 setting, the index candle with a body of 80% or more traded in its direction at ITM20, so nothing went to the test. Its choice also missed the luck bar: the shifted searches reached its score in 9 of 195 cases ($p = 0.051$), and its gain came almost entirely in the first half of the tuning period. What the search adds is comparative (Table 10). The 15:50 candle led, 15:54 came second, every other minute scored below both, and two minutes had no eligible setting at all. Ranked afterwards over all of 2021–2026, all fifteen leading settings were 15:50 candles traded in their direction, most of them at ITM20; the first, both candles at 60% or more, made +$34.74 a day ($t = 2.71$, 505 trades). That ranking reuses days already seen and is not a test.

Table 10: The best eligible setting for each candle, 2021–2023.

The highest-scoring eligible setting among those that read the given candle, of 387 eligible among 7,128. Score is the mean profit per day averaged with the same setting one cut above and below; $/day is the setting's own mean over 625 sessions. One contract at the ask in the snapshot that opens the next minute, held to settlement, $0.09 in fees. "Both" requires the index and SPY candles to pass the cut in the same direction. Pre-registered.

Candle	Source	Body	Side	Strike	Score	$/day	Trades
15:45	Index	$\geq 80\%$	With	ITM20	+$21.46	+$27.65	166
15:46	Index	$\geq 40\%$	With	ITM5	+$1.48	+$6.81	430
15:47	Index	$\geq 70\%$	With	ITM5	−$0.19	+$4.25	235
15:48	Index	$\geq 40\%$	With	ATM	+$19.30	+$19.41	416
15:49	Index	$\geq 90\%$	With	ATM	+$5.22	+$7.75	104
15:50	**Index**	**$\geq 80\%$**	**With**	**itm20**	**+$37.46**	**+$39.42**	**193**
15:51	No eligible setting						
15:52	Both	$\geq 50\%$	Against	ITM5	+$12.00	+$13.31	237
15:53	Index	$\geq 80\%$	With	ATM	+$23.40	+$23.69	177
15:54	Index	$\geq 60\%$	With	ITM5	+$33.55	+$32.43	299
15:55	No eligible setting						

7.3 Other studies

Four further studies, reported in full in the Internet Appendix, found nothing that improved the rule.

Early exits. Under the calibrated model, six stop-loss and take-profit rules on the index path from 15:51 to 15:58 each earned less on average than holding, at every strike and in every model variant; stops cut the worst loss by more than half but cost money on average. The signal is a small drift over nine minutes, many adverse moves reverse before the close, and every early sale pays the spread again.

Intraday momentum and dealer gamma. An adapted replication of the market intraday momentum regression, on the CFD over 2013–2026, found a positive but insignificant slope ($t = 1.02$), mildly positive only in 2013–2015 and about zero on 2021–2026 ($t = -0.18$), the only period outside the samples of Gao et al. (2018) and Baltussen et al. (2021). This does not show that the published results are wrong, since samples and instruments differ. No filter that kept the candle trade only when it agreed with the day's earlier move improved it: trades that agreed did better per trade, but the trades a filter skipped were still profitable on average. A net-gamma proxy built from the open interest of every SPX and SPXW option, assuming that dealers are long every call and short every put, did not pass either test: under model prices the candle trade did better on negative-gamma days, but the difference missed its bar ($p = 0.029$ against 0.025) and was absent at real asks on 2021–2026.

Smaller contracts. The SPX contract cost about $1,100 on 2021–2026, so the same signal was traded on XSP (Mini-SPX) and SPY options about a tenth the size. Neither was confirmed: XSP lost $49.66 a trade ($t = -5.48$ per day) and SPY broke even. The SPX trade's edge was about six cents per dollar of premium, and both smaller contracts cost more than that to trade; XSP's median quoted spread was 31% of the option's price, against 5% for SPX on the same days.

Forward evidence. From 21 October to 6 November 2026, the strong-body candle fired on seven of thirteen sessions and was right on six. Only four of those days have real one-minute option prices, and four trades support no conclusion.

8 Why 15:50?

The evidence points to one minute. A scan of 41 features selected the 15:50 candle, and on 2021–2023 the re-tuning over 2,178 settings and the search over 7,128 both chose a 15:50 candle (tuning on 2013–2020, when spreads were wide and only two settings were eligible, had favoured a 15:42 candle). A strong 15:50 candle is followed by an index move in its direction whether it is read from a CFD, from the index or from SPY. We did not set out to explain the effect, but the timing of the closing auction offers a natural hypothesis, and one pre-registered study tests its most direct prediction.

8.1 The closing-auction hypothesis

From 15:50 the NYSE's regulatory closing imbalance is public and Nasdaq's early imbalance indicator is disseminated, and the NYSE accepts new market-on-close and limit-on-close orders only on the side that offsets the imbalance. The 15:50 one-minute candle is therefore the first full minute in which the market can react to the published closing imbalance. A candle with a strong body is a minute in which prices moved one way with little retracement, as might happen when a large imbalance is published. If liquidity providers then absorb the imbalance gradually up to the auction, the index would keep drifting in the same direction for the remaining minutes, which is the pattern we find: a move of about two index points over nine minutes, in the candle's direction, only when the candle has a strong body. This reading fits the evidence of Bogousslavsky and Muravyev (2023) that imbalance information contributes to price discovery at the close, and it would explain why the effect requires a one-sided minute rather than a large one.

The hypothesis has to contend with two facts. The signal was already present in 2013–2020, for most of which the NYSE's cut-off and mandatory publication came five minutes earlier, at 15:45. Until late 2018 the only imbalance first published at 15:50 was Nasdaq's, and for a few months after that no exchange began publishing at 15:50, so the NYSE's schedule cannot be the whole story. And a strong body on the index candle alone, without the CFD and SPY candles, carried no gain, which suggests that the futures-linked instruments react first. The exchanges' imbalance messages allow a direct test, which we report next. Whether the effect concentrates on days with large imbalances, and whether it changed when the NYSE moved its publication from 15:45 to 15:50 in April 2019, a few months after Nasdaq had moved its own from 15:50 to 15:55, we leave to future work.

8.2 A test with the published imbalances

If traders react to the published imbalance during the 15:50 minute, the candle should be right more often on days when it points the same way as that imbalance. A pre-registered study (plan locked 2026-11-10 19:37 UTC, one registered run) tested this prediction on the rule's 644 signal days of 2021–2026. For each year it took the 50 largest stocks in SPY at the previous year-end, from the fund's N-PORT reports to the SEC (between 51% and 63% of its net assets), and bought from Databento the imbalance messages that each stock's listing exchange sent from 15:50. From each stock's first closing-imbalance record received in the 15:50 minute it summed the signed dollar imbalance, $D = \sum_i s_i Q_i p_i$, with $s_i = +1$ for a buy imbalance, -1 for a sell imbalance and 0 for none, Q_i the imbalance in shares and p_i the reference price. A day agrees if the sign of D is the candle's direction. The primary test compares the rule's hit rate on agreeing and disagreeing days, $d = h_a - h_d$, and passes if $d > 0$ and $t \geq 2.0$. By the plan's power calculation it could detect a true d of about 11 to 13 points, a large concentration of the edge on agreeing days, not a modest one. All 50 stocks had a usable record on every day, so every signal day was classified.

The candle and the imbalance usually pointed the same way, on 469 of the 644 days (72.8%). But agreement did not make the candle more accurate (Table 11): it was right on 55.9% of agreeing

Table 11: Signal days that agree and disagree with the 15:50 closing imbalance, by year.

Signal days of the rule (a 15:50 body of 60% or more), 2021–2026. A day agrees when the sign of the summed dollar imbalance of the 50 largest stocks, from each stock's first closing-imbalance record received in the 15:50 minute, equals the candle's direction. Every signal day was classified, so no day is unclassified. Hit rate: share of days on which the CFD moved from the 15:51 open to the 15:59 close in the candle's direction, with its Wilson 95% interval in percent; bp: the mean of that move in the candle's direction, in basis points. Over all days the difference in hit rates is −5.9 points ($t = -1.35$), with a 20-day block-bootstrap 95% interval from −14.8 to +4.0 points.

		Agreeing days			Disagreeing days		
Year	Days	n	Hit rate (95% CI)	bp	n	Hit rate (95% CI)	bp
2021	111	92	62.0% (52–71)	4.67	19	78.9% (57–91)	5.12
2022	105	76	57.9% (47–68)	4.00	29	69.0% (51–83)	3.88
2023	113	73	53.4% (42–64)	0.33	40	55.0% (40–69)	1.18
2024	106	69	59.4% (48–70)	2.59	37	43.2% (29–59)	−1.90
2025	117	91	53.8% (44–64)	1.39	26	69.2% (50–83)	3.97
2026	92	68	47.1% (36–59)	0.96	24	70.8% (51–85)	4.29
All	644	469	55.9% (51–60)	2.41	175	61.7% (54–69)	2.24

days and 61.7% of disagreeing days, $d = -5.9$ points ($t = -1.35$), with a block-bootstrap interval from −14.8 to +4.0 points, and disagreeing days had the higher hit rate in five of the six years, all but 2024. The imbalance direction alone was right on 51.1% of days. The robustness checks fixed in the plan agree: the difference in hit rates stayed negative with the latest record of the minute, with NYSE or Nasdaq stocks alone and within year and direction, and the differences in the average move and in the option P&L were small against their noise ($t = 0.13$ and 0.79; Internet Appendix). The prediction was therefore not confirmed. The test has limited power, and a failure does not refute the auction explanation: the price may adjust within the 15:50 candle itself, imbalances change after their first publication, and 50 stocks are only part of the index. The high agreement is an association, not a cause, since the candle and the imbalance can reflect the same closing orders. What the test does show is that the rule's edge was not concentrated on the days on which its candle agreed with the imbalance first published at 15:50.

9 Conclusion

One end-of-day candle survived a pre-registered, out-of-sample test: the 15:50 one-minute candle with a strong body, traded in its own direction to the close. It is a known effect in a sharper form, and it has faded since 2023. Expressed with a deep in-the-money 0DTE option, bought at the quoted ask and held to settlement, it made money on 2021–2026, though less than a pricing model implied. The bid-ask spread is the decisive cost: it took the whole edge before 2021, and a limit order at the mid does not recover it, because it misses the best days. Nothing tried afterwards, from re-tuned settings and real-time exit rules to momentum filters and smaller contracts, improved on the rule, and a search over the other closing candles chose the 15:50 candle again.

The confirmation holds for the candle as the backtests read it, from a CFD that tracks the index. Read from the SPX index candle under a plan locked in advance, the rule earned less and did not meet its criterion; read from the SPY candle it met a retrospective one; and on every source the gain came on sessions on which all three candles showed a strong body. A pre-registered test found that the candle usually agrees with the closing-auction imbalance published at 15:50 but is no more accurate when it does; with limited power, that leaves the mechanism open. The fair summary is a small, execution-sensitive effect that is worth continuing to measure, not an established edge. The study's limits follow from its design: the holdout was nominal rather than blind, later studies reused the 2021–2026 outcomes, the fills are proxies for what an order would have paid, and the

luck diagnostic is approximate.

Three steps would sharpen the evidence. Settings suggested by the searches, such as stricter 15:50 cuts and the agreement of several candles, should be tested only on days after November 2026, under new registered plans. Closing-imbalance data could test the rest of the hypothesis of Section 8.1: whether the effect depends on the size of the imbalance, and whether it changed when the NYSE moved its publication to 15:50 in 2019. And forward records of actual order fills should replace the fill proxies, since they alone can settle how the trade should be executed.

Data and code availability. The analysis code, the pre-registration plans, their hash ledger and the Internet Appendix are public at `https://github.com/kv1514/spx-closing-candle-study`. The OPRA data are licensed from Databento and are not redistributed.

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